Strong generalisation: a theorem about an unused constant promotes to a universal

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the constant-freshness condition appears verbatim in both statement and the middle proof line; the explain-environment discussion preceding the theorem is outside the span.

Editorial note. The label thm:strong-generalization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Deduction theorem. Same chapter, the deduction theorem (source line 656). Lines 1131 and 1134: entered through truth and exited back to bare derivability.
- Universal generalisation on axioms/theorems (gen rule for conditional with true antecedent). First-order axioms of this chapter: from |- top -> A(c) infer |- top -> forall x A(x) when c occurs in neither the assumptions nor top. Line 1132-1133 is the single inference step between the two deduction-theorem moves.

Theorem 0.1. *If c is !!aconstant not occurring in Γ or $!A(x)$ and $\Gamma \vdash !A(c)$, then $\Gamma \vdash \forall[x][!A(x)]$.*

Proof. By the deduction theorem, $\Gamma \vdash \top \to !A(c)$. Since c does not occur in Γ or $\top$, we get $\Gamma \vdash \top \to \forall[x][!A(x)]$. By the deduction theorem again, $\Gamma \vdash \forall[x][!A(x)]$. □